

Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 6

Percentiles, Subpopulation Analysis, and Functions of Estimates

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Estimation of Percentiles

- Analysts are often interested in estimating more than just the mean to describe the distribution of a variable in a population
- Example: 95th percentile of PSA levels in a national sample of men over age 40
- Example: Median total assets (a very skewed variable) for rural homes
- Example: Quartiles of systolic blood pressure for African-American females

The Ungrouped Estimation Method

- First, Define the population CDF for a variable:

$$F(x) = \frac{\sum_{i=1}^N I(y_i \leq x)}{N}$$

- A weighted estimator of this CDF can be written as follows:

$$\hat{F}(x) = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i} I(y_{h\alpha i} \leq x)}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{\alpha}} w_{h\alpha i}}$$

The Ungrouped Estimation Method

- The q -th quantile for a variable y is the smallest value of the variable such that the estimated CDF is greater than or equal to q ($q = 0.25, 0.5, 0.75$, etc.)
- First order sample values from smallest to largest, $x_{(1)}, \dots, x_{(n)}$, and then find the value of j such that

$$\hat{F}(x_j) \leq q < \hat{F}(x_{j+1}).$$

The Ungrouped Estimation Method

- Then, a weighted estimate of the q -th population quantile, X_q , is computed as follows:

$$\hat{X}_q = x_j + \frac{q - \hat{F}(x_j)}{\hat{F}(x_{j+1}) - \hat{F}(x_j)} (x_{j+1} - x_j)$$

- Taylor Series Linearization (Binder, 1991) approaches can be used to estimate the variance of this estimate, but BRR approaches are recommended (Kovar, Rao, and Wu, 1988)

Software for Design-based Estimation of Percentiles

- SUDAAN: Taylor Series Linearization (TSL)
- WesVar: Balanced Repeated Replication (recommended)
- SAS: Woodruff's Method (uses TSL)
- Procedures enabling appropriate computation of standard errors and confidence intervals (using BRR) are not readily implemented in the current versions of Stata and SPSS
- Stata users can install the user-written *epctile* command (`findit epctile, net install epctile.pkg`)

Estimation of Percentiles: Total Household Wealth (2012 HRS)

	SAS(TSL)		SAS(BRR)		Stata(TSL)	
Percentile	$\hat{Q}_p$	$se(\hat{Q}_p)$	$\hat{Q}_p$	$se(\hat{Q}_p)$	$\hat{Q}_p$	$se(\hat{Q}_p)$
Q ₂₅	\$21,953	\$2,215	\$21,953	\$2,287	\$22,000	\$2,200
Q ₅₀ (Median)	\$141,907	\$7,754	\$141,907	\$7,761	\$142,000	\$8,000
Q ₇₅	\$439,965	\$18,652	\$439,965	\$19,940	\$440,000	\$18,869

Stata code:

```
svyset secu [pweight = nwgthh], strata(stratum)
epctile H11ATOTA, percentiles(25 50 75) subpop(if nfinr==1) svy
```

Subpopulation Analysis

- Examples of Subpopulations:
 - Men, women
 - Age groups
 - Disease groups (asthma sufferers)
 - Voters
 - Home Owners
 - Persons with income >\$20,000

“Unconditional” vs. “Conditional” Analysis

- Cochran (1977), West et al. (2008) – “Conditional” analysis “conditions” on observed subpopulation sizes as though they were fixed. Results from using “if”, “by”, or “where” statements when analyzing the data.
- Conditional analysis is OK for simple random samples...
- ...but not necessarily for stratified samples:
 - Distribution of subpopulation cases, $m(h)$, to strata $h=1, \dots, H$ is a **random variable**. *Rarely* fixed.
 - Correct if the subpopulation of interest is used to define explicit strata, e.g., Census Region in a multi-stage national sample of U.S. households, or gender in a stratified (by sex) sample of men and women in a University student body.

“Unconditional” vs. “Conditional” Analysis

- “Unconditional” analysis treats stratum subpopulation sizes as a random variable, $m(h)$, $h=1, \dots, H$
 - Variability of subpopulation across strata and to clusters within strata (including $m(hi)=0$) must be reflected in the variance estimation.
- Stata subpop() and SAS DOMAIN keywords/options ensure that this variability is reflected in standard errors of estimates.

“Unconditional” vs. “Conditional” Analysis

- Incorrect conditional analyses can also result in the case where only one sampling error computation unit (SECU) is detected within a stratum by the software, preventing appropriate variance estimation!
- Sampling error calculation models based on combining strata attempt to prevent this from happening...
- ...but caution is still needed. When in doubt, use the unconditional approach!

Unconditional Analyses in Stata

- `over(varname)` option for command
 - *varname* is a categorical variable
 - Analyses (correct) will be replicated for each level of the categorical *varname*
- `subpop(varname)` for command
 - *varname* is a user-generated 0,1 variable where 1 indicates membership in subclass
 - applies to all svy procedures

Unconditional Analyses in Stata

- The Stata subpop() feature always produces the correct result.
- Stata examines the design distribution (by strata and clusters) of the subpopulation. If a stratum contains 0 cases, that stratum and its clusters do not contribute to degrees of freedom, i.e., $DF = \# \text{ clusters} - \# \text{ of strata}$.

Functions of Estimates: Stata svy: mean Mean Household Assets by Education (2012 HRS)

- `gen finr = 1`
- `replace finr = 0 if kfinr != 1`
- `svyset secu [pweight = nwgthh], strata(stratum)`
- `svy, subpop(finr): mean h11atota, over(edcat)`

Education of Head	Stata Label	$\bar{y}_w$	$se(\bar{y}_w)$	$CI_{.95}(\bar{y}_w)$
0-11 yrs	1	\$122,089	\$10,595	(\$100,863 , \$143,314)
12 yrs	2	\$259,027	\$9,802	(\$239,390, \$278,664)
13-15 yrs	3	\$336,308	\$17,201	(\$301,849, \$370,768)
16+ yrs	4	\$834,141	\$46,478	(\$741,035, \$927,247)

Sampling Errors for Functions of Estimates: Differences of Subpopulation Means

$$\text{var}\left(\sum_{j=1}^J a_j \hat{\theta}_j\right) = \sum_{j=1}^J a_j^2 \text{var}(\hat{\theta}_j) + 2 \cdot \sum_{j=1}^{J-1} \sum_{k>j}^K a_j a_k \cdot \text{cov}(\hat{\theta}_j, \hat{\theta}_k)$$

where: a_j, a_k are any chosen constants.

Example:

$$\text{var}(\bar{y}_{\text{sub1}} - \bar{y}_{\text{sub2}}) = \text{var}(\bar{y}_{\text{sub1}}) + \text{var}(\bar{y}_{\text{sub2}}) - 2 \text{cov}(\bar{y}_{\text{sub1}}, \bar{y}_{\text{sub2}})$$

where: $\bar{y}_{\text{sub1}}, \bar{y}_{\text{sub2}}$ are estimates of the mean of y for
two subclasses.

Sampling Errors for Functions of Estimates: Complex Sample Designs

- Sampling errors for statistics that are functions of survey estimates require computations of variances and covariances of estimates. These should be stored by the software in a variance/covariance matrix.
- In complex sample designs, the covariance terms for subpopulation estimates are often positive due to the clustering (non-independence) in the design.
- This is true even when the subpopulations are distinct (male and female). The covariance will be zero if the subpopulations are defined by distinct design strata (e.g., Northeast vs. South Region).

Stata Example: Compute Subpopulation Means

```
. svy, vce(linearized): mean numadl, over(arthrtis)
```

```
1: arthrtis = 1
```

```
2: arthrtis = 2
```

		Mean	Std. Err.	[95% Conf. Interval]	
numadl					
	1	.9487063	.0356663	.8780096	1.019403
	2	.3940189	.0279537	.3386098	.449428

Stata Example (2): Compute the standard error for the difference of the subpopulation means.

```
. lincom [numadl]1-[numadl]2
```

```
( 1)  [numadl]1 - [numadl]2 = 0
```

	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
(1)	.5546874	.0408345	13.58	0.000	.4737463	.6356285

Note: Difference
in two estimates
from previous
slide!

Stata Example: Display Variance/Covariance Matrix for the Two Subpopulation Means

```
. estat vce
```

Covariance matrix of coefficients of mean model

		numadl	
e(V)		1	2

numadl			
	1	.00127208	
	2	.00019302	.00078141

Difference of Means: Stata lincom post-estimation Total Assets by Education Level of Head: 2012 HRS

*Command immediately following svy: mean command from
Slide 15:*

lincom [h11atota]1 - [h11atota]4

Education of Head	$\bar{y}_{0-11}$ $\bar{y}_{16+}$	$se(\bar{y}_{0-11} - \bar{y}_{16+})$	$CI_{.95}(\bar{y}_{0-11} - \bar{y}_{16+})$
0-11 vs. 16+	-\$712,052	\$48,886	(-\$809,983, -\$614,122)

Note: Difference
in two estimates
from Slide 15.

Stata Example: Display Variance-Covariance Matrix for the Subpopulation Means

. estat vce

Covariance matrix of coefficients of mean model

Subpopulation	1	2	3	4
1	1.123×10^8	0.257×10^8	0.564×10^8	-0.587×10^8
2		0.961×10^8	0.640×10^8	0.194×10^8
3			2.959×10^8	1.572×10^8
4				2.160×10^{10}

Difference of Means: Hand Computation

Total Assets by Education Level of Head: 2012 HRS

$$\begin{aligned} se(\bar{y}_{0-11} - \bar{y}_{16+}) &= \sqrt{var(\bar{y}_{0-11}) + var(\bar{y}_{16+}) - 2 \times cov(\bar{y}_{0-11}, \bar{y}_{16+})} \\ &= \sqrt{[1.123 + 21.600 - 2 \times (-0.587)] \times 10^8} \\ &= \$48,886 \end{aligned}$$

Example: Estimating a Difference of Means over Time
Total Assets: HRS 2004 vs. 2006

```
gen weight = jwghh  
replace weight = kwghh if year == 2006  
gen finr04 = 1 if (year==2004 & jfinr==1)  
gen finr06 = 1 if (year==2006 & kfinr==1)  
gen finr0406 = 1 if finr04==1 | finr06==1  
svyset secu [pweight = weight], strata(stratum)  
svy, subpop(finr0406): mean totassets, over(year)  
lincom [totassets]2004 - [totassets]2006
```

Contrast	$\bar{y}_{2004} - \bar{y}_{2006}$	$se(\bar{y}_{2004} - \bar{y}_{2006})$	$CI_{.95}(\bar{y}_{2004} - \bar{y}_{2006})$
2004 vs. 2006	-\$115,526	\$20,025	(-\$155,642, -\$75,411)

Example: SAS Difference in Means using PROC SURVEYMEANS and PROC SURVEYREG

- Use of PROC SURVEYMEANS to calculate means/se
- Subsequent use of PROC SURVEYREG with a LSMEANS / diff statement for test if means are significantly different

PROC SURVEYMEANS Output

```
proc surveymeans data=one;
strata sestrat ; cluster seclustr ;
weight ncsrwtlg ;
var bmi ;
domain ed4cat;
run ;
```

The SURVEYMEANS Procedure

Data Summary

Number of Strata	42
Number of Clusters	84
Number of Observations	9282
Number of Observations Used	5692
Number of Obs with Nonpositive Weights	3590
Sum of Weights	5692.00048

Statistics

Variable	Label	N	Mean	Std Error of Mean	95% CL for Mean	
bmi	Body Mass Index	5099	27.160539	0.129700	26.8987936	27.4222842

Domain Analysis: 1=0-11 Years 2=12 Years 3=13-15 Years 4=16+ Years

1=0-11 Years 2=12 Years 3=13-15 Years 4=16+ Years	Variable	Label	N	Mean	Std Error of Mean	95% CL for Mean	
1	bmi	Body Mass Index	767	27.371506	0.309041	26.7478353	27.9951760
2	bmi	Body Mass Index	1521	27.705417	0.182695	27.3367223	28.0741111
3	bmi	Body Mass Index	1528	27.161737	0.228553	26.7004982	27.6229761
4	bmi	Body Mass Index	1283	26.265901	0.169481	25.9238745	26.6079268

PROC SURVEYREG Output

```
proc surveyreg data=one;
strata sestrat ; cluster seclustr ;
weight ncsrwltg ;
class ed4cat ;
model bmi=ed4cat / solution ;
lsmeans ed4cat / diff;
run ;
```

ED4CAT Least Squares Means					
1=0-11 Years 2=12 Years 3=13-15 Years 4=16+ Years	Estimate	Standard Error	DF	t Value	Pr > t
1	27.3715	0.3091	42	88.54	<.0001
2	27.7054	0.1827	42	151.60	<.0001
3	27.1617	0.2286	42	118.81	<.0001
4	26.2659	0.1695	42	154.93	<.0001

Differences of ED4CAT Least Squares Means						
1=0-11 Years 2=12 Years 3=13-15 Years 4=16+ Years	1=0-11 Years 2=12 Years 3=13-15 Years 4=16+ Years	Estimate	Standard Error	DF	t Value	Pr > t
1	2	-0.3339	0.3368	42	-0.99	0.3272
1	3	0.2098	0.3235	42	0.65	0.5202
1	4	1.1056	0.3221	42	3.43	0.0014
2	3	0.5437	0.3186	42	1.71	0.0953
2	4	1.4395	0.2427	42	5.93	<.0001
3	4	0.8958	0.2717	42	3.30	0.0020